Lab3

1. Exercise 1

#Source Code

```
students = {
    112345: "Annie",
    112346: "Lucy",
    112347: "Lily",
    112348: "Moly",
    112349: "Jesica"
}
students[112350] = "Jennie"
students[112351] = "Jessie"
students[112347] = "Lillian"

student_id_to_retrieve = 112346
retrieved_student = students.get(student_id_to_retrieve)
print(f"Student ID {student_id_to_retrieve} corresponds to: {retrieved_student}")

student_id_to_delete = 112345
del students[student_id_to_delete]
print(f"Student ID {student_id_to_delete} has been deleted.")

non_existing_student_id = 112345
try:
    print(f"Trying to retrieve student ID {non_existing_student_id}: {students[non_existing_student_id]}")
except KeyError:
    print(f"Key Error: Student ID {non_existing_student_id} not found.")
```

#Result:

```
Student ID 112346 corresponds to: Lucy
Student ID 112345 has been deleted.
Key Error: Student ID 112345 not found.
```

2. Exercise2

#Source Code

```
student = {
   112345 : "Annie",
   112346 : "Lucy",
   112347 : "Lily",
   112348 : "Moly",
   112349 : "Jesica"
}

ID = student.keys()
print("📌 Student IDs:", list(ID))

name = student.values()
print("📖 Student Names: ", list(name))

ID_and_name = student.items()
print("🎓 Student Directory: ", list(ID_and_name))

Searching_student = student.get(112345)
print("🔍 Searching_student: ",(Searching_student))

remove_student = student.pop(112345)
print(f"🗑️  Remove Student: ", (remove_student))
```

updated_student = student.clear()

print(f" 📝 Student directory has been reset: ", (student))

#Result:

```
📌 Student IDs: [112345, 112346, 112347, 112348, 112349]
📖 Student Names:  ['Annie', 'Lucy', 'Lily', 'Moly', 'Jesica']
🗺️ Student Directory:  [(112345, 'Annie'), (112346, 'Lucy'), (112347, 'Lily'), (112348, 'Moly'), (112349, 'Jesica')]
🔍 Searching_student:  Annie
🗑️ Remove Student:  Annie
📝 Student directory has been reset:  {}
```

3. Exercise3

#Source Code

```
import sys

contact_book = {}

contact_book ={

   "Luccy" : '0123456789',

   "Lily" : '0983467489',

   "Moly" : '0987654321',

   "Jescica" : '0882122123',

   "Jenn" : '0882222243'

}

print(f"Please search your contact here")

search_contact = input("Enter contact's name: ")

try:

   if search_contact in contact_book:

      print(f" 📞 Contact Found! {search_contact}: {contact_book[search_contact]}")

   else:

      print("Contact not found")

except KeyError:

   print("Contact not found")

choice = input("Do you want to update your contact here?(yes/no): ").strip().lower()

if choice == "yes":
```

```
    print("Please update your contact here")
else:
    print("Goodbye")
    sys.exit()
update_contact = input("Enter add contact's name: ")
update_number = input("Enter add contact's number: ")
contact_book[update_contact] = update_number
updated_contact_book = contact_book | contact_book
print(f"Your new cantact book")
for key, value in updated_contact_book.items():
  print(f"{key}:{value}")
choice = input("Do you want to delete contact from your contact book?(yes/no): ").strip().lower()
if choice == "yes":
    print("Please delete your contact by enter contact's name below ")
else:
    print("Thank You! Goodbye.")
    sys.exit()
delete_name = input("Enter delete contact's name: \n")
del updated_contact_book[delete_name]
print(f"Your new cantact book after deleted contact")
for key, value in updated_contact_book.items():
  print(f"{key}:{value}")
```

#Result:

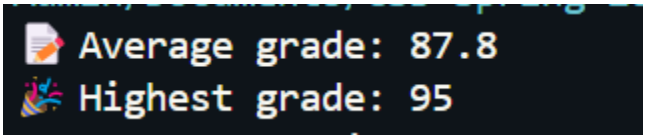
```
Please search your contact here
Enter contact's name: Lily
📞 Contact Found! Lily: 0983467489
Do you want to update your contact here?(yes/no): yes
Please update your contact here
Enter add contact's name: Nia
Enter add contact's number: 012767678
Your new cantact book
Luccy:0123456789
Lily:0983467489
Moly:0987654321
Jescica:0882122123
Jenn:0882222243
Nia:012767678
Do you want to delete contact from your contact book?(yes/no): no
Thank You! Goodbye.
```

4. Exercise 4

#Source code

```
gradebook = {
  "John": 85,
  "Mary": 92,
  "Paul": 78,
  "Anna": 89,
  "Steve": 95
}
def average_grade(gradebook):
  total = 0
  for student in gradebook:
    total += gradebook[student]
  return total / len(gradebook)
print(f"📝 Average grade: {average_grade(gradebook)}")
def highest_grade(gradebook):
  highest = 0
  for student in gradebook:
    if gradebook[student] > highest:
      highest = gradebook[student]
  return highest
print(f"🎉 Highest grade: {highest_grade(gradebook)}")
```

#Result:



5. Exercise 5

#Source code

```
def word_frequency(sentence):
```

```
    word = sentence.split()
    word_count = {}
    for w in word:
        if w in word_count:
            word_count[w] += 1
        else:
            word_count[w] = 1
    return word_count
sentence = input("Enter a sentence: ")
result = word_frequency(sentence)
print("Word frequency:")
for word, count in result.items():
    print(f"{word}: {count}")
```

 # Result

```
Enter a sentence: Do it until you make it
Word frequency:
Do: 1
it: 2
until: 1
you: 1
make: 1
```

6. Challenge Exercise

 # Source code

```
courses = {
    "Math": {
        "Chesda": 98,
        "Luccy": 35,
        "Loire": 89,
    },
    "English": {
        "Jen": 95,
```

```
        "Moly":98,
        "Kaily": 92,
    },
    "Science": {
        "Jesica": 90,
        "Lily": 100,
        "Lyka": 45,
    },
    "History": {
        "Lesly": 90,
        "Veha": 100,
        "Zoe": 98,
    },
    "Geography": {
        "Vella": 90,
        "Chigi": 100,
        "Wansy": 98,
    },
}

def average_grade(course_name):
    if course_name in courses:
        grades = courses[course_name].values()
        return sum(grades) / len(grades)
    else:
        return None
def highest_grade(course_name):
    if course_name in courses:
        students = courses[course_name]
```

```
        return max(students, key=students.get)
    else:
        return None
def passing_students(course_name):
    if course_name in courses:
        return {student for student, grade in courses[course_name].items() if grade >= 50}
    else:
        return set()
def failed_students(course_name):
    if course_name in courses:
        return {student for student, grade in courses[course_name].items() if grade < 50}
    else:
        return set()
if __name__ == "__main__":
    print("Average Grade in Math:", average_grade("Math"))
    print("Highest Grade in History:", highest_grade("History"))
    print("Passing Students in Science:", passing_students("Science"))
    print("Failed Students in Math:", failed_students("Math"))
```

 # Result:

```
Average Grade in Math: 74.0
Highest Grade in History: Veha
Passing Students in Science: {'Lily', 'Jesica'}
Failed Students in Math: {'Luccy'}
```